IRIDIUM CERTUS 9604 · HYBRID IOT

Map the gap before you map the hardware.

A practical readiness guide for cellular-plus-satellite connected products.

REVIEW EDITION · NOT A COVERAGE ASSESSMENT

September 2026

01 · START WITH OPERATING REALITY

Where does the product actually go?

The useful coverage map is the one your device, asset or worker experiences—not a general market assumption.

01
Map movement
Record routes, operating regions, depots, fields, sites, dwell points and seasonal changes.

02
Mark uncertainty
Separate verified network evidence from known gaps, anecdotal reports and locations still to test.

03
Name the consequence
Describe what the operation cannot see or do when the preferred communications path is unavailable.

04
Select a test route
Choose one representative scenario for prototype and field validation. Do not design from the best case.

Working rule: label every coverage claim as observed, provider-supplied, inferred or unverified. A map is evidence to test—not a guarantee.

Readiness input	Your working evidence
Representative route or site	
Known / suspected coverage boundary	
Operational consequence	

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02 · RANK THE MESSAGES

What needs to travel now?

A stronger hybrid design starts by separating decision-changing exceptions from routine data.

Message class	Questions to answer
Routine telemetry	Can it store, batch or retry? What delay is acceptable?
Threshold event	What condition changes an operating decision? What is the smallest useful payload?
Location exception	Which movement or boundary requires attention? Does it need acknowledgement?
Safety escalation	Who responds, how is delivery state shown and what happens when communications are unavailable?
Configuration / downlink	How urgent is it? What control and validation prevent unintended changes?

Design principle: use the most appropriate path for each message class. “Transmit everything everywhere” is rarely a sound power, service-cost or operational strategy.

Urgency

How long can the message wait?

Consequence

What decision changes if it arrives?

Evidence

How will delivery and acknowledgement be observed?

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03 · DEFINE HYBRID BEHAVIOUR

The module enables the paths. Your system defines the logic.

The Iridium Certus 9604 combines LTE-M, Iridium Short Burst Data and GNSS. It does not remove the need to design routing, retry and fallback behaviour.

1

Choose a preferred path

Define when cellular is attempted and what evidence determines availability.

2

Set retry and timeout rules

Make timing explicit for each message class; avoid a single rule for all traffic.

3

Define satellite escalation

Specify which events, payloads and operating conditions justify the satellite path.

4

Design the unavailable state

Store locally, continue safe local behaviour and make link limitations visible to users and operations.

5

Prove it in a field test

Retain route, radio, message, acknowledgement, power and outcome evidence.

> **Claim control:** do not describe the 9604 as automatically switching networks by itself. Iridium's product information states that the subsystems can be independently controlled; switching and routing logic are implementation choices.

04 · DESIGN THE COMPLETE DEVICE

Connectivity is one subsystem.

A viable hybrid product joins radio behaviour with physical, electrical, firmware, commercial and operating constraints.

☐ Power

Budget for searches, retries, transmission, sleep, GNSS and worst-case operating conditions.

☐ Antennas

Review cellular and satellite RF paths, placement, ground plane, isolation, sky view and installation variability.

☐ Enclosure

Account for size, materials, ingress protection, temperature, mounting and service access.

☐ Firmware

Define state machine, queues, retry, acknowledgements, diagnostics, updates and safe failure behaviour.

☐ Payload

Prioritise compact, decision-useful messages and confirm the applicable network limits.

☐ Service model

Model traffic, activation, support, expected exceptions and the commercial consequence of each path.

☐ Certification

Confirm product, regional, carrier, RF, safety and industry obligations for the intended deployment.

☐ Operations

Make message state, delayed data, loss of link and escalation visible to the people who act.

Define “working” before the field test.

A green process is not the outcome. Retain evidence that the intended message travelled, was understood and supported the operating decision.

Evidence layer	What to retain
Environment	Route/site, time, weather or obstruction notes, installation and antenna state.
Device	Firmware version, power state, network state, queue and retry behaviour.
Message	Class, timestamp, payload size, selected path, attempts and acknowledgement.
Platform	Ingestion, processing, visible state, alerts and operator action.
Outcome	Did the intended decision occur? What failed, degraded or remained unknown?

Pass condition
Write a measurable result for each scenario before testing.

Rollback condition
Define when the design returns to the last verified behaviour.

Limitations
Record what the route, duration and sample cannot prove.

Owner
Name who accepts technical, operational and safety evidence.

06 · READINESS WORKSHEET

Bring these seven inputs.

Input	Ready when…	Status / owner
1 · Operating map	A representative route or site is documented.	
2 · Coverage evidence	Known, inferred and unverified areas are distinguished.	
3 · Message profile	Message classes, size, urgency and acknowledgement are defined.	
4 · Device constraints	Power, antenna, enclosure and environmental constraints are available.	
5 · Behaviour	Preferred path, retry, escalation and unavailable state are explicit.	
6 · Commercial model	Traffic assumptions, service costs and support ownership are modelled.	
7 · Test plan	Scenarios, pass conditions, evidence and approval owners are named.	

Suggested first review: choose one representative scenario, one critical message and one operational owner. A bounded test produces better evidence than a broad technology demonstration.

07 · NEXT STEP

Turn the coverage concern into a design conversation.

Bring the operating map, critical messages and device constraints. We will help assess whether a cellular-plus-satellite architecture fits the problem and what needs proving next.

Cellular where you can. Satellite where you must.

Start at m2mone.com.au/hybrid-iot-readiness/

Start a Hybrid IoT design conversation →

Evidence and limitations

This guide is a campaign-owned diagnostic aid. It is not a radio survey, network availability guarantee, safety certification, product certification or final architecture. Technical claims must be checked against the current product documentation and intended operating region before release or design commitment.

Primary sources observed 9 September 2026
Iridium Certus 9604 product information: iridium.com/9604
Iridium commercial availability announcement, 23 June 2026: investor.iridium.com
M2M One Hybrid IoT readiness page: m2mone.com.au/hybrid-iot-readiness/